TQUK Functional Skills Qualification in Maths at Level 2

Examination Past Paper 5

Please complete the details below using black or blue ink. Use BLOCK CAPITALS.

Learner Name: _____

Learner Number: _____

Date: _____

Centre Name: _____

Instructions:

- read each question **carefully**
- answer **all** questions
- write your answers **clearly** in the spaces provided
- **check** your answers.

Information:

- this examination has **two** sections. These are clearly labelled
- you are **not allowed** to use a calculator for Section A
- you **are allowed** to use a calculator for Section B
- the **maximum** mark for this examination is 60
- the marks available for each question are shown in **bold** beneath each question.

OFFICIAL MFS MODEL SOLUTION · BENCHMARK EXEMPLARY PASS

60 / 60 Marks (100% Pass · Full Marks)

· Assessment: Past Paper 5 (Non-Calculator & Calculator)

· Complete stepped calculations, methods & mark scheme codes [MI/AI]

· Exam Board: TQUK (Training Qualifications UK · Ofqual Regulated)

· Pass standard: 35 / 60 marks (Achieved: 60 / 60 marks)

Verified by Tutor Abdul (Senior Assessor) · MasterFunctionalSkills.co.uk

Total marks	60		

MASTER FUNCTIONAL SKILLS

ONLINE LEARNING PLATFORM & ACCREDITED COURSES

Pearson Edexcel & Open Awards Accredited Suite · Entry 3, Level 1 & Level 2

Pass Your Functional Skills Exams With Confidence

100% Pass Guarantee | Dyslexia-Friendly Learning | 1-to-1 Expert Tutor Support

Master Functional Skills provides an adaptive, dyslexia-friendly online learning platform tailored for adult learners, apprentices, and NHS healthcare staff. Everything you need to revise, practice, and pass is at your fingertips.

1. Smart Diagnostic Initial Assessment

Calculate your exact working level and pinpoint knowledge gaps instantly

- **Exact Grade Calculation:** Adaptive assessments calculate your exact level across all topics.
- **Personalised Study Pathway:** Skip what you know; focus 100% on the topics needed to pass.
- **Real-Time Exam Readiness:** Predictive tracker confirms when you are 100% ready to book.

2. Comprehensive Video Lessons & Worked Solutions

Bite-sized visual explainers covering every single Maths & English curriculum standard

- **Step-by-Step Video Lessons:** Master percentages, algebra, perimeter, ratio, and reverse fractions.
- **Multi-Ink Handwriting Models:** Visual 3-colour methods showing working, carries, and examiner tips.
- **Tutor Shortcuts & Golden Rules:** Proven non-calculator mental arithmetic hacks and BIDMAS tricks.

3. Full Mock Exams & Real-Time Performance Analytics

Practice in exam conditions and watch your scores and confidence climb

- **Authentic Exam Papers:** Timed Section A & Section B papers formatted exactly like Pearson Edexcel.
- **Instant Topic Breakdown:** View percentage scores across Number, Measure/Shape, and Data.
- **Historical Score Tracking:** Monitor past attempts to ensure continuous improvement before exam day.

Ready to Master Your Functional Skills & Pass First Time?

Join thousands of healthcare staff, apprentices, and adult learners across the UK.

Instant 24/7 access on desktop, tablet, and mobile with no software to install.

START LEARNING TODAY AT

MasterFunctionalSkills.co.uk

 © Training Qualifications UK 2023 All Rights Reserved

Section A: Non-calculator



There are **15 marks** available in this section.
You must **not** use a calculator in this section.
You will have **30 minutes** to complete this section.

1. Work out 9.486 – 0.52

Calculate: 14.616 - 5.65 = 8.966

Answer 8.966 ✓ [1 mark]

[1]

2. Work out 4 × 2.125

Calculate: 68 ÷ 8 = 8.5

Answer 8.5 ✓ [1 mark]

[1]

Past paper

Questions continue on the following page

 © Training Qualifications UK 2023 All Rights Reserved

3. Work out:

$$8\frac{1}{4} + 2\frac{1}{5}$$

Give your answer as a mixed number.

Common denominator of 4 and 5 is 20:

8 1/4 = 8 5/20 and 2 1/5 = 2 4/20

Total = (8 + 2) + (5/20 + 4/20) = 10 9/20

Answer	10 9/20 ✓ [2 marks]

[2]

4. Work out $100 + 5^2 \times 2$

BIDMAS: Powers and Multiplication before Subtraction

3² = 9 --> 9 x 2 = 18 --> 25 - 18 = 7

Answer	7 ✓ [2 marks]

[2]

Past paper

© Training Qualifications UK 2023 All Rights Reserved

5. Dale is an estate agent and sells two properties.

These are the sale prices:

House A:	£825 750

House B:	£390 595

Dale thinks that the total sale price for both properties is more than £1 215 000

Is Dale correct?

Show how you decide.

Step 1: Calculate 8.5% of £340

Calculation: 340 × 0.085 = £28.90

Answer £28.9 ✓ [2 marks]

[2]

Past paper

Questions continue on the following page

Master Functional Skills pack - TQUK Mathematics Level 2 Past Paper 5 Solutions © Training Qualifications UK 2023 All Rights Reserved

6. A concrete block has:

- mass = 4800 grams
- volume = 2000 cm^3

Work out the density of the concrete block.

Total ratio parts = 3 + 2 = 5 parts

1 part = 120 ÷ 5 = 24 --> Larger share = 24 × 3 = 72

Answer 72 *[2 marks]*

[2]

7. The distance between Taylor's house and Kai's house on a map is 8 centimetres (cm).

The scale on the map is 1 : 50 000

Work out the actual distance between the two houses.

Give your answer in kilometres (km).

Map scale: 1 cm = 50 000 cm = 500 m = 0.5 km

8 cm × 50 000 = 400 000 cm = 4000 m = 4 km

Answer 4 *[2 marks]* **km**

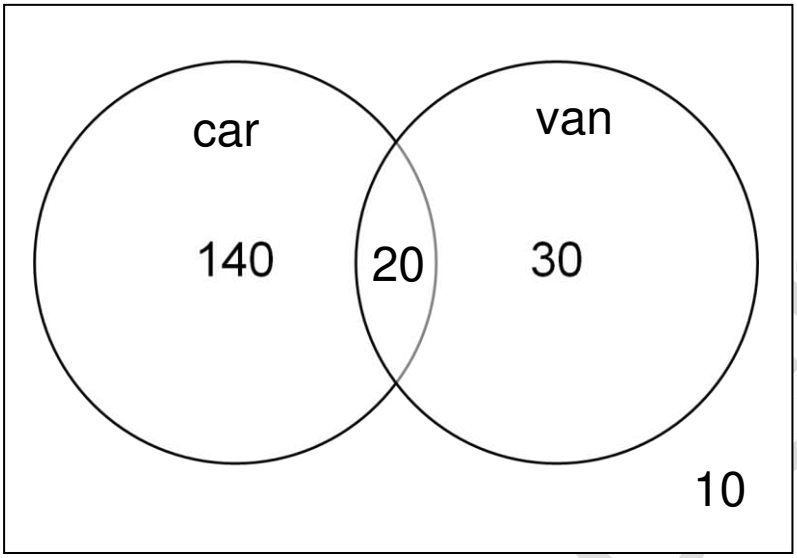
[2]

Past paper

© Training Qualifications UK 2023 All Rights Reserved

8. Lei asks a group of people whether they own a car or a van.

The results are shown in this Venn diagram:



One of these people is chosen at random.

What is the probability that the person chosen does **not** own a car or a van?

Give your answer as a decimal **and** a percentage.

Probability = 10 / 200 = 1 / 20

Decimal: 10 ÷ 200 = 0.05 | Percentage: 0.05 × 100 = 5%

Answer	a) Decimal: 0.05	b) Percentage: 5%

[3 marks]

[3]

End of Section A.

Past paper

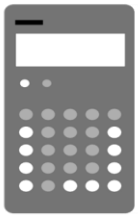
© Training Qualifications UK 2023 All Rights Reserved

Section B begins on Page 7.

Past paper

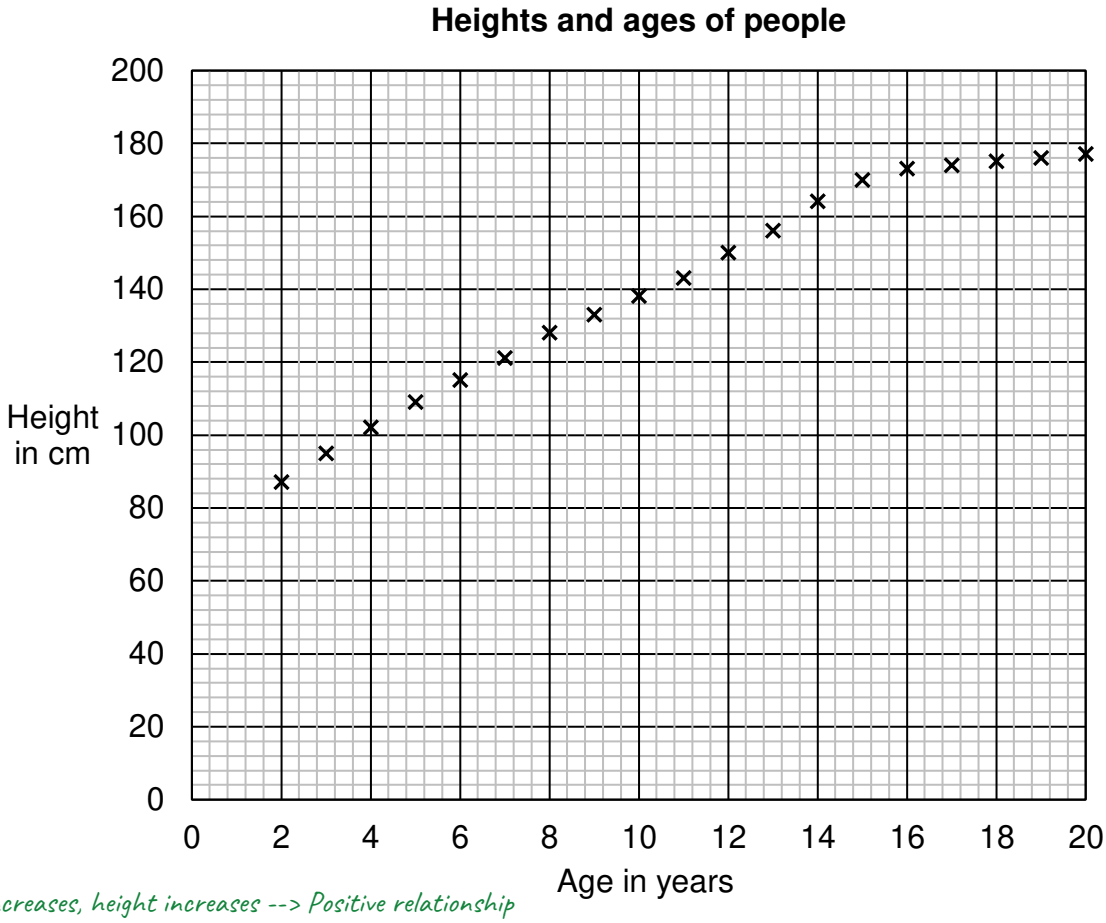
 © Training Qualifications UK 2023 All Rights Reserved

Section B: Calculator



There are **45 marks** available in this section.
You **can** use a basic calculator in this section.
You will have **90 minutes** to complete this section.

1. What type of correlation does the scatter graph show?



As age increases, height increases --> Positive relationship

a) Positive correlation	**b)** Negative correlation
c) No correlation	**d)** Weak correlation
Answer	Positive relation [1 mark]

[1]

Questions continue on the following page

© Training Qualifications UK 2023 All Rights Reserved

2. Write down the mode of these numbers:

1000 2000 1200 1500 1200 1400 1375

Mode is the most frequently occurring value in the list:

1200 appears twice; all other values appear once.

Answer 1200 *[1 mark]*

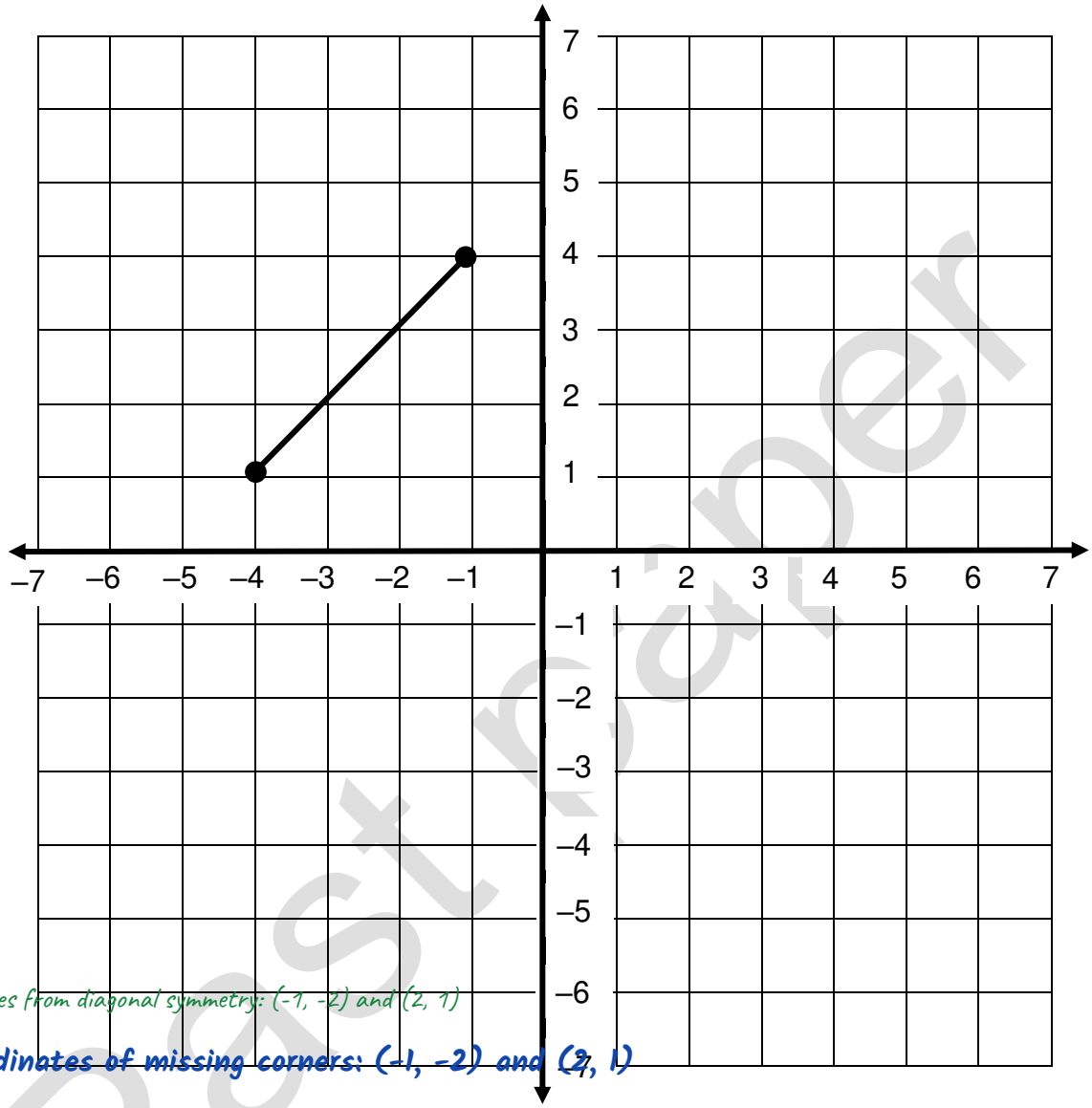
[1]

Past paper

 © Training Qualifications UK 2023 All Rights Reserved

3. Riley wants to put a square vegetable patch in a garden.

One side of the vegetable patch is shown on the grid below:



Vertices from diagonal symmetry: (-1, -2) and (2, 1)

Coordinates of missing corners: (-1, -2) and (2, 1)

Write down possible coordinates for the other **two** corners of the square.

Answer	(-1, -2) and (2, 1) [2 marks] (,) and (,)

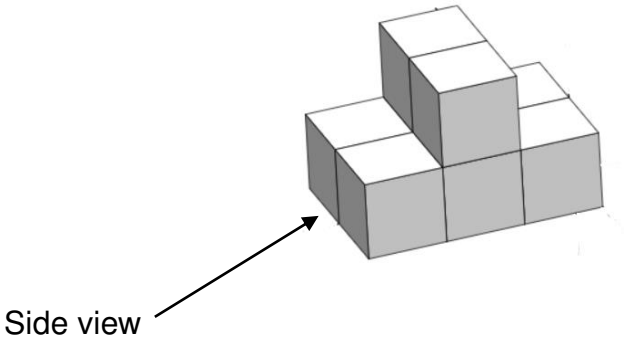
[2]

Past Paper

Questions continue on the following page

 © Training Qualifications UK 2023 All Rights Reserved

4. Draw the plan view and side view of the shape:



Plan view **Side view**

Plan View

Side View

[2]

Plan view: square/rectangle seen from above.

Side view: elevation seen from the side.

[2 marks]

© Training Qualifications UK 2023 All Rights Reserved

5. Write:

$\frac{1}{16}$

a) as a decimal

b) as a percentage.

Fraction 1/16 = 1 ÷ 16 = 0.0625 | Percentage = 6.25%

Answer	a) Decimal: 0.0625 ✓	b) Percentage: 6.25% [2 marks]

[2]

6. Jessie reads reviews for a film online.

210 of the reviewers would recommend the film.

30 of the reviewers would **not** recommend the film.

What percentage of the total number of reviewers would recommend the film?

Total visitors = 210 + 30 = 240

Percentage = (210 ÷ 240) × 100 = 87.5%

Answer	87.5 ✓ [2 marks] %

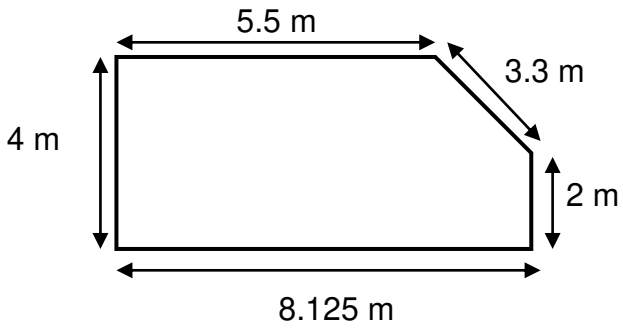
[2]

Questions continue on the following page

Past paper

© Training Qualifications UK 2023 All Rights Reserved

7. Sam needs to put a fence around the garden shown below:



Not drawn to scale

Fence panels are 2.6 metres wide.

The fence panels can be cut to size after Sam buys them.

How many fence panels does Sam need to buy?

Perimeter = 4 + 5.5 + 3.3 + 2 + 8.125 = 22.925 m

Panels needed = 22.925 ÷ 2.6 = 8.817... panels --> Rounds up to 9 panels

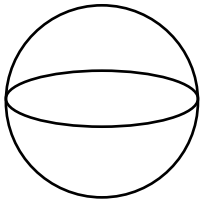
Answer 9 [3 marks] **panels**

[3]

Past paper

 © Training Qualifications UK 2023 All Rights Reserved

8. Stevie needs to estimate the surface area of a sphere.



Stevie knows:

- the radius of the sphere is 8 cm
- $\pi = 3.14159$

Stevie rounds π to the nearest whole number to **estimate** the surface area of the sphere.

What answer should Stevie get?

Use:

$$A = 4\pi r^2$$

Where:
A = surface area of sphere
r = radius of sphere

Show your working.

Formula: Surface Area = 4 × π × r^2

Estimate using π ≈ 3: 4 × 3 × 8^2 = 12 × 64 = 768 cm^2

(Using π = 3.14: 4 × 3.14 × 64 = 803.84 cm^2)

Answer 768 ✓ [3 marks]	**cm^2**

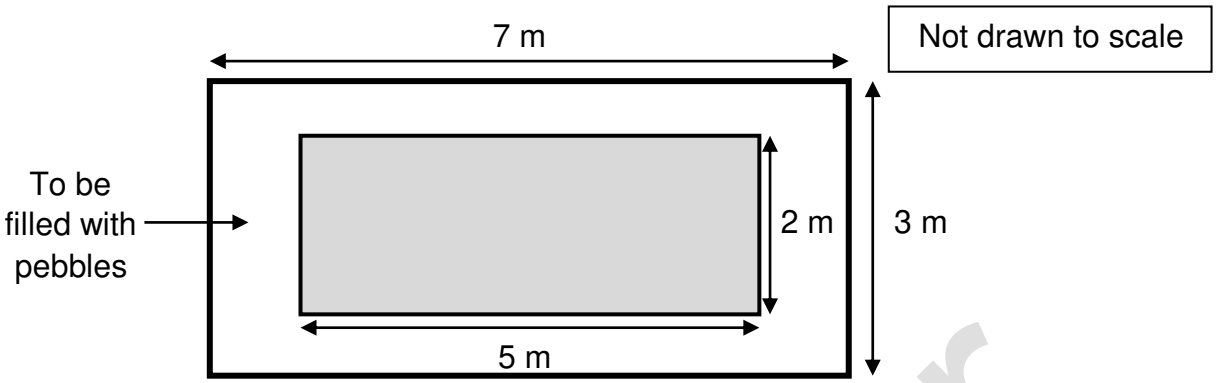
[3]

Past paper

Questions continue on the following page

 © Training Qualifications UK 2023 All Rights Reserved

9. Parker wants to fill the path around a fishpond with grey and white pebbles to a depth of 0.06 metres (m).



Parker uses this formula to estimate the total amount of pebbles needed:

(Area of larger rectangle – area of smaller rectangle) multiplied by depth of pebbles

Parker reduces the total amount by 38% to estimate the amount of white pebbles required.

What answer should Parker get?

Base Area = (7 × 3) - (5 × 2) = 21 - 10 = 11 m²

Total Volume = 11 m² × 0.06 m = 0.66 m³

Pebbles (100% - 38% = 62%): 0.66 × 0.62 = 0.4092 m³

Answer 0.409 ✓ *[3 marks]* **m³**

[3]

Past paper

 © Training Qualifications UK 2023 All Rights Reserved

10. Hayden cycles 15 miles in 50 minutes.

Hayden thinks it is more than 17 miles per hour (mph).

Is Hayden correct?

Show how you decide.

Time in hours = 50 ÷ 60 = 0.833... hours

Speed = Distance ÷ Time = 15 ÷ (50/60) = 18 mph

Comparison: 18 mph > 17 mph. Therefore Yes, she cycled faster.

Answer	Yes (18 mph > 17 ✓) [3 marks] mph

[3]

Past paper

Questions continue on the following page

 © Training Qualifications UK 2023 All Rights Reserved

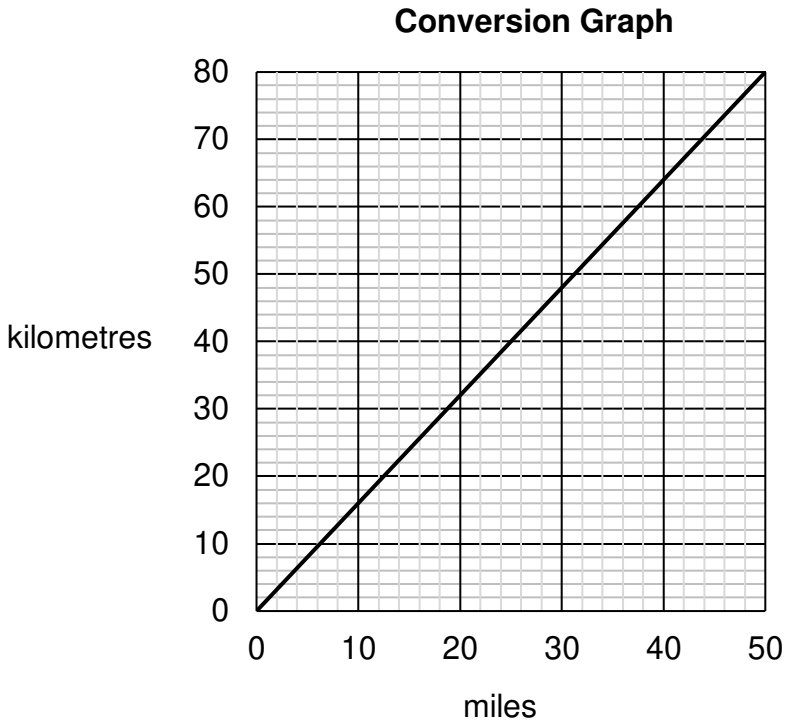
11. Kass is planning a road trip.

The total distance is 4000 kilometres (km).

Kass only wants to drive a maximum of 180 miles each day.

How many whole days will Kass need to complete the 4000 km?

Use this graph to convert between kilometres and miles.



Conversion from graph: 80 km = 50 miles (8 km = 5 miles)

4000 km = (4000 ÷ 8) × 5 = 2500 miles

Days = 2500 ÷ 180 = 13.888... days --> Rounds up to 14 days

Answer 14 [4 marks] **days**

[4]

 © Training Qualifications UK 2023 All Rights Reserved

12. Jude needs to order 200 sandwiches for a party.

Jude has these instructions:

- order cheese, ham and chicken sandwiches in the ratio 5 : 2 : 1

Complete the shopping list for Jude.

Shopping list	
Type of sandwich	**Number of sandwiches to order**
Cheese	
Ham	
Chicken	
Total	

Ratio Cheese : Ham : Chicken = 5 : 2 : 1 → Total = 8 parts

1 part = 200 ÷ 8 = 25 sandwiches

Cheese = 5 × 25 = 125 | Ham = 2 × 25 = 50 | Chicken = 1 × 25 = 25

125

50 ✓ [4 marks]

25

[4]

Past paper

Questions continue on the following page

 © Training Qualifications UK 2023 All Rights Reserved

13. Nia and Charlie are both taxi drivers.

The table shows how far each drove last week:

	Monday	Tuesday	Wednesday	Thursday	Friday
Nia	140 miles	86 miles	138 miles	110 miles	96 miles
Charlie	76 miles	120 miles	124 miles	60 miles	130 miles

Their manager thinks that, on average, Nia drove further each day than Charlie.

Charlie disagrees.

Show that **both** could be correct.

Show your working.

Morning: Mean = 114, Median = 110

Afternoon: Mean = 102, Median = 120

Comment: If using the mean, Manager is correct (114 > 102).

If using the median, Charlie is correct (120 > 110).

Answer	Manager (Mean) / Charlie (Median) [4 marks]

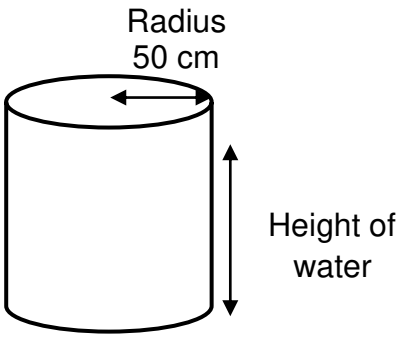
[4]

Past paper

© Training Qualifications UK 2023 All Rights Reserved

14. Harper works at an aquarium.

Harper needs to fill this cylindrical fish tank until it is 72% full.



Not drawn to scale

Harper knows that the tank will hold 981 250 cubic centimetres (cm^3) of water when it is full.

To what height does Harper need to fill the tank?

Use $\pi = 3.14$

Water volume = 981 250 × 0.72 = 706 500 cm^3

Cylinder cross-section = $\pi \times r^2$ = 3.14 × 50^2 = 7850 cm^2

Height = 706 500 ÷ 7850 = 90 cm

Answer 90 [5 marks] **cm**

[5]

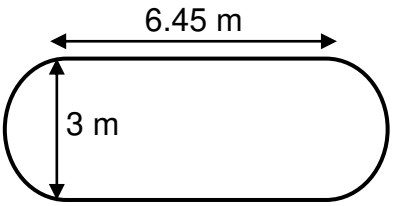
Questions continue on the following page

Past paper

© Training Qualifications UK 2023 All Rights Reserved

15. Dylan wants to estimate how much it will cost to carpet a section of floor.

The section of floor is made up of a rectangle with a semi-circle on each end.



Not drawn to scale

Dylan finds these carpets for sale:

Carpet	A	B	C	D	E
Price per square metre (m^2)	£22.50	£15.75	£18.00	£24.95	£15.75

Dylan uses the modal price to estimate how much the total cost will be.

How much does Dylan estimate it will cost to carpet the section of floor?

Give your answer to the nearest pound (£).

Use π = 3.14

Area = 2 semi-circles + rectangle Area = 2 × 7.065

Circle Area = $\pi \times 1.5^2 = 3.14 \times 2.25 = 7.065\ m^2$

Rectangle Area = $6.45 \times 3 = 19.35\ m^2$

Answer box is on the following page

Past paper

 © Training Qualifications UK 2023 All Rights Reserved

Total Area = 7.065 + 19.35 = 26.415 m^2

Modal price of seed = £15.75 per m^2

Total cost = 26.415 m^2 × £15.75 = £416.036... --> £416

Answer	£ 416	[6 marks]

[6]

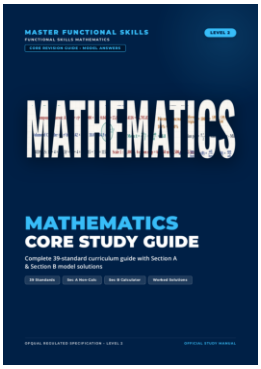
Past paper

MASTER FUNCTIONAL SKILLS STORE

OFFICIAL REVISION GUIDES, WORKBOOKS & FLASHCARDS

Written by accredited specialist tutors to match the exact 2026 Pearson Edexcel standards

Everything You Need To Revise, Practice & Pass Your Maths Level 2 Exam



Core Curriculum Manual

Maths Level 2
Core Study Guide

- Full official curriculum coverage
- Section A Non-Calc & Sec B Calc
- Worked models & golden rules
- Dyslexia-friendly visual format

Spiral Bound Print & PDF

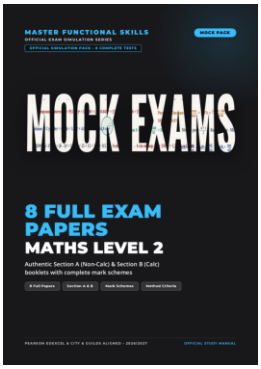


1,000+ Practice Drills

Maths Level 2
Companion Workbook

- 1,000+ exam-style questions
- Progressive difficulty drills
- Full worked step-by-step key
- Targeted fix for weak areas

Spiral Bound Print & PDF



Realistic Mock Papers

Maths Level 2
Official Mock Packs

- Full mock exam paper sets
- Authentic Pearson Edexcel style
- Detailed mark schemes included
- Grade boundaries & pass marks

Exam Booklet Pack & PDF



Pocket Flashcards A6

Maths Level 2
Pocket Revision Cards

- Compact A6 cards for quick study
- Essential formulas & conversions
- Dual-sided question & answer
- High-yield mental arithmetic rules

Boxed Card Deck & Print PDF

Order Official Study Guides & Revision Cards Online

Free Next-Day UK Tracked Shipping · Instant PDF Downloads Included

Trusted by NHS trusts, apprentices & adult learners nationwide.

VISIT THE ONLINE STORE

MasterFunctionalSkills.co.uk